DUNSTAN - PREDICTING CUSTOMER CLAIMS

Anuja Kanmathukalam Varghese< kaan23gi@student.ju.se>
Ayana Stephen Puthur< puay23mt@student.ju.se>
Ayham Hanna<haay20uj@student.ju.se>
Bincy Annamma Saji< puay23mt@student.ju.se>
Essa Alshehada< ales18zn@student.ju.se>
Likhith Bedara Agrahara Venkateshanurthy<beli23fr@student.ju.se>
Muhammed Rizwan< ramo23ad@student.ju.se>

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Introduction

The following report contains information regarding a project's main goal which appears to be enhancing the efficiency and effectiveness of the veterinary care insurance program. The key objectives and expected results suggest a multi-faceted approach aimed at understanding and addressing the factors contributing to increased claims costs. The project was arranged by Science Park in Jönköping who met Dunstan and managed to organize the project.

Our project topic is based on Dunstan Insurance Company. Dunstan is a specialized insurance company that focuses entirely on the unique needs of those involved in the horse and farm community.

This report is about making horse and farm insurance better at Dunstan. Dunstan is a special insurance company that helps people in the horse and farm world with their insurance needs. They want to make sure the insurance they offer is smart and fits the specific needs of people living and working on farms with horses.

As the injured animals has gone up by 17% at Dunstan, the project focuses on understanding customer behavior and claims patterns. The goal is to predict the claims beforehand which could be the base for pricing model adapted to customer profiles, based on behavior analysis.

Business Case

In our project, the main focus is on addressing the challenges posed by the excessively high increase in veterinary care costs, particularly due to a 17% increase in costs from veterinarians during the year. The increased number of injuries and higher injury costs per diagnosis, coupled with claims costs in relation to earned premiums is affecting the overall profitability of the business. The current pricing strategy aligns with competitor prices, but there is a need to adjust the weighting of existing ones to improve the overall pricing model.

The company has implemented an incentive program aimed at promoting education and increased knowledge among horse and farm owners to prevent claims. The effectiveness of this program needs evaluation – specifically, whether claims are predominantly due to ignorance or if they are a result of factors beyond control. Understanding customer behavior and profiles can provide insights into the patterns associated with claims. Some of the main challenges in our project are:

1. Identifying Patterns and Correlations

   Understanding the patterns between customer behavior and profiles and how they are related to claims categories can be complex.

2. Predicting Claims Events

   Developing a reliable predictive model for claims events involves advanced analytics. Factors such as customer behavior and other external variables must be taken into consideration to make accurate predictions.

3. Individualized Pricing

   Implementing individualized pricing based on claim history is one of the challenges.

4. Behavior Analysis and Training

   Assessing whether claims are due to ignorance or bad luck involves understanding customer behavior. Designing effective training programs and incentives to increase knowledge and prevent claims requires a deep understanding of customer motivations.

5. Excel Model Improvement

   Enhancing the Excel model for follow-up and analysis requires a clear understanding of the data inputs, formulas, and overall usability. It involves integrating new data sources, improving data visualization, and ensuring accuracy in calculations.

To achieve the long-term goals of lowering claims costs, increasing profitability, promoting sustainability, and enhancing customer satisfaction, we are focusing on three main aspects.

- Understanding customer behavior and claims patterns – our aim is to identify patterns between customer behavior, customer profiles, environmental factors, education levels, and specific claims areas/categories.
- Refining the pricing model - Exploring the possibility of pricing more individually based on behavior analysis, claims patterns, and customer profiles.
- Improving the Excel Model for Follow-Up and Analysis - The current Excel model is expected to undergo improvement to better facilitate follow-up and analysis.

Expected outcome in our project are:

· Lower claims costs and increased profitability for company and customer satisfaction through targeted claim prevention efforts.

· Enhanced sustainability for humans, horses, and farms by addressing the root causes of claims.

Data description

Once the business understanding is obtained, the next steps was to explore the data that was provided. The data provided to work was two excel sheets. The first and foremost sheet is regarding the premium pricing details and premium types which consisted of all the policies which was registered to company. This sheet also included details of the clients which can be useful to analyze the customer behavior who took the policies. Table 1 illustrates the name and type of the data in premium. The latter sheet contained was for all the claims information which was reported to the company and it is shown in Table 2. In this step the focus was spend was to have an understanding what does each data excel was accounted. This was done by having analyzing the data types, total count, unique count and null values. As the most of the data was in Swedish, we also had taken some extra time in converting into English by referring the Table 3 and 4. One of the main observation/challenges from this step was that the ratio of customer who had claim and who hadn't claims was unbalanced as the amount of data in claims data is very less when compare to premium. This will affect the model building stage which we discuss in the further portions of the report.

Column name		
Försäkrad	CALC_Objektid	CALC_AntalObjekt Perkund
Utpris	CALC_AntalDagarUtanför...	CALC_AntalObjektPerkund
Förfallen (utpris)	CALC_RadAktiv	Aktuell älder häst
Intjänat (utpris)	CALC_Rad_Aktiv_Annu_Ko...	Total skadekostnad
Produkt	Calc_Ålder_häst	Totalt utbetalt skada
Moment	CALC Ålderskatekori	Utgående skadereserv
Anslutsdatum	Total skadekostnad	Skadeår
Förmedlare	Totalt utbetalt skada	Anslutningsår
Försäkringsnummer	Utgående skadereserv	Antal skador
Hästens namn (värde)	CALC_FlaggaFörstaObjekt	Antal moment
Period start	CALC_AntalObjektPerkund	Antal moment ej nollskada
Period slut	CALC_AntalObjektPerkund	Ålder häst skadetillfälle
Annullerad	Aktuell älder häst	
CALC_Kundid	CALC_Flagga FörstaObjekt...	

Table [1] Shows the names of each column in premium data.

Column name	
Skadenummer	Intjänat (utpris)
Skadedatum	Produkt
Registrerad	Moment
Status	Anslutsdatum
Typ	Förmedlare
Diagnoskod	Försäkringsnummer
Försäkring Självrisk veterinärvård varde	Hästens namn (värde)
Utbetalningar	Period start
Försäkring Försäkringsbelopp	Period slut
Försäkrad	Annullerad
Utpris	CALC_Kundid
Förfallen (utpris)	

Table [2] Shows the name and datatype of each column in claims data.

Data preparation

After having data understanding the next major step which we focused on in the project is data preparation. This was the stage where we spent most of our time on this project. This phase involved cleaning and transformation of the data so that it can be useful for the data analysis and modelling. The steps involved in the data cleaning and transformation are described briefly in the next few steps. During the data preparation we had ensured that the data was ready for analysis and model building.

Imputation

The missing values are one of the problems we encounter when we tried to prepare the data for the model. The reason for the missing values may vary from human errors, or the data collection way to privacy concerns. In our work flow we had missing values in different stages like after reading the data and after joining the different data together. The way we handled the problem vary depending on which stages. In the claims data we started data cleaning by examining the rows with the maximum number of null values. After analyzing this we concluded that Deductible Vet care, Deductible Vet care value, Indemnity Outstanding, Indemnity Paid, Sum insured, Total Incurred had the majority of values being null. Among these columns the sum insured column had only 14% data as non- null values and since it cannot be replaced by any imputation, we had removed the specific column. During the data understanding phase, with the company meeting we have confirmed that the deductible with null values can be replaced with 0, as 0 indicates that the customer is not paying any deductible for the policy. The number of null values in the rest columns were not as much as the other columns so we did not perform any data cleaning on those columns, but in the Column selection columns we have excluded them.

Figure 1 below shows the statistics of the number of missing values in each column for claims data.

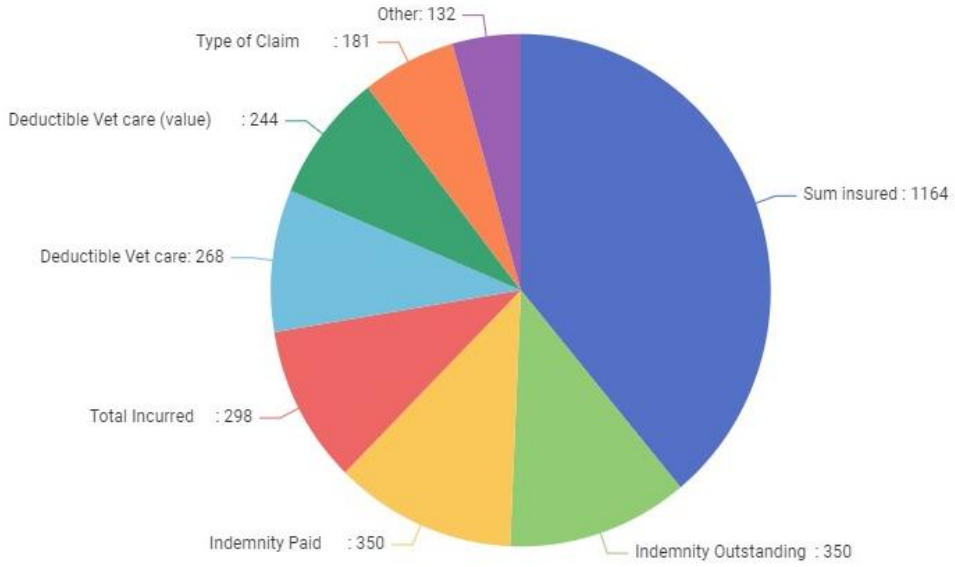

Fig [1] Pie chart describing missing values in claims data.

After completing data cleaning on the claims data, we had applied the same process for premium data.

The columns which gave details about the horse and customer also had some null values. The race and the horse's name had some null values that indicated that either the horse had no race and name or the customer had been missed to specify it to the company. This missing value was dealt by substituting the value as *No Race* and *No Name* so that the null value doesn't affect our model. Even for the customer details like age and gender there existed some null-values, this was because the type of customer is business (not a person). We had our analysis on the basis of customer type separately, hence these columns were not a part of any data-cleaning.

Below is the figure 2 which is the statistics of number of missing values in each column for premium data.

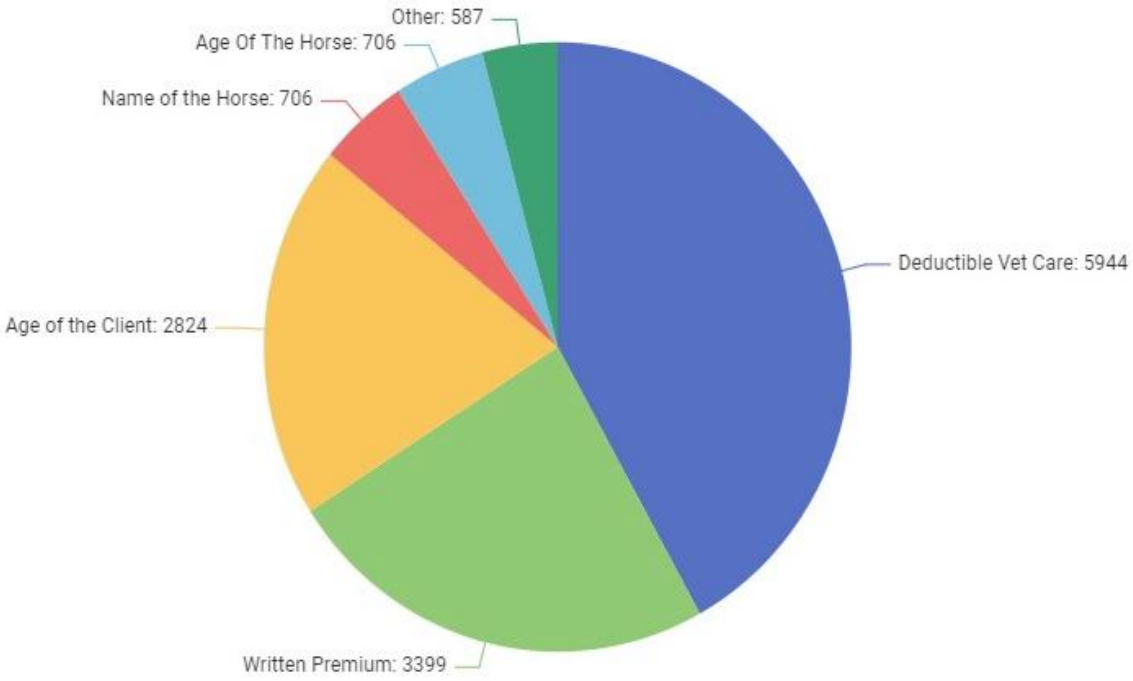

Fig [2] Pie chart of missing values in premium data.

Figure 3 is the snapshot of the data after handling the columns name of horse and race.

Row ID	S Type of Policy - Insurance element	S Policy Number	S Name of the Horse	S Race
Row0	Premium Veterinärvård	209187	Föl undan Lap Dance	Svenskt Varmblod (SWB)
Row1	Premium Liv & Användbarhet	209188	Föl undan Lap Dance	Svenskt Varmblod (SWB)
Row2	Grund Livförsäkring	209189	Föl undan Lap Dance	Svenskt Varmblod (SWB)
Row3	Premium Veterinärvård	210045	Bonmahon Diana	Irländsk Sporthäst
Row4	Grund Livförsäkring	210046	Bonmahon Diana	Irländsk Sporthäst
Row5	Breeding Veterinärvård	217118	Actrice	Svenskt Varmblod (SWB)
Row6	Dunstan Safe Start Premium Foster	217787	No Name	No Race
Row7	Dunstan Safe Start Premium Föl	217788	No Name	No Race
Row8	Premium Veterinärvård	221404	Skjioni	Islandshäst
Row9	Premium Veterinärvård	226686	Lilly off Beauty	Svenskt Varmblod (SWB)
Row10	Premium Liv & Användbarhet	226687	Lilly off Beauty	Svenskt Varmblod (SWB)
Row11	Grund Livförsäkring	226688	Lilly off Beauty	Svenskt Varmblod (SWB)
Row12	Premium Veterinärvård	244602	Sarinosa	Zangersheide
Row13	Premium Veterinärvård	244603	Cococera	Zangersheide
Row14	Breeding Veterinärvård	244604	Lap Dance	Svenskt Varmblod (SWB)
Row15	Premium Liv & Användbarhet	244605	Sarinosa	Zangersheide
Row16	Premium Liv & Användbarhet	244606	Cococera	Zangersheide
Row17	Breeding Liv & Användbarhet	244607	Lap Dance	Svenskt Varmblod (SWB)
Row18	Premium Veterinärvård	80767	Power of Bologna	Varmblodstravare
Row19	Grund Livförsäkring	80768	Power of Bologna	Varmblodstravare
Row20	Premium Veterinärvård	94177	Qlara	Svenskt Varmblod (SWB)
Row21	Grund Livförsäkring	94178	Qlara	Svenskt Varmblod (SWB)
Row22	Special Veterinärvård	155579	Miss M Star	Engelsk Fullblod
Row23	Grund Livförsäkring	155580	Miss M Star	Engelsk Fullblod
Row24	Dunstan Safe Start Föl	159727	No Name	Övriga Halvblod
Row25	Premium Veterinärvård	172738	Semlan	Tysk Ridponny

Figure [3] Data after handling the name and race of the horse.

Data Joining

Once the data is cleaned, we had joined the claims and premium data for having a single data table with customers who had raised the claim and who had not raised the claim.
The merging was done two times, in the first time the key for joining was UW Year and Policy Number as this was the unique values for each policy.
Before joining we had encountered an issue which is, that the policy number in claims data had the suffix '-0' at the end of every of policy number. This issue was dealt by excluding the suffix from the end of every policy number.

Below figure 4 highlights the policy number before and after handling

Policy number	New Policy Number
130272-0	130272
73072-0	73072
237161-0	237161
210188-0	210188
191131-0	191131
185245-0	185245
202471-0	202471
145319-0	145319
213383-0	213383
35589-0	35589
182412-0	182412
128376-0	128376
61158-0	61158
208012-0	208012
124382-0	124382
174144-0	174144
159233-0	159233

Fig [4] Policy Number before and after handling.

As the business had customers with insurance for other products other than the horse (Eg: house insurance, horse cart insurance) we had to filter out the rows related to horse products only as the aim was to study the analysis on horse insurance.

The premium and claims data had some columns for which we did not do any cleaning as we excluded these columns for our further analysis. This reasoning for this is mentioned in the column selection chapter.

Column creation

The data provided does not contain many columns that directly represent consumer behaviour, but it contains information about the number of claims, the customer, and the start date for the insurance that the customers have.
From the start date we created a new column called ‘registrationPeriod’ to be used to create new columns. By extracting data from this information and ‘registrationPeriod’’ we create new columns like 'claims per person', 'claims per year' and 'claims per person and per year’, to provide additional context that might help statistical and machine learning models to make better predictions or classifications. The new columns offer more depth and understanding of the underlying processes or customer’s behaviour in the data. For instance, 'claims per person and per year' might show patterns or trends that are not apparent when looking at the total number of claims alone.

The other data we could create was the target data which is if the customer has raised claim or not. In order to get that data, we had divided the total count of damage number by itself, by doing this we got new columns ‘0’ and ‘non’ which indicates that whether the customer had raised a claim and customer did not raise any claims respectively.

We had also created other columns and this was done by grouping the data according to a criterion. The 2 different criteria which we used for grouping are based on customer’s number and horse’s name and by customer id to create new columns that represent customer’s behaviour like total number of unique customers who have filed a claim.
.

Column Selection

This phase was repeated over at least three times, each repetition was performed through three steps, which are choosing columns candidates, applying columns selection methods. The methods employed in columns selection are Forward columns Selection, Backward columns Selection and Genetic Algorithm columns Selection. The goal was to focus more on the columns that contribute significantly to our predictive models.

For the selection process, we relied on Cohen's Kappa value, where Cohen's Kappa is a statistical measure used to assess the agreement between the chosen columns during the columns selection process that helps to understand how well the selected columns align with each other and contribute to the accuracy of the predictive models. The value of Cohen's Kappa ranges from 0 to 1 and it is better if the value is nearest to 1. Cohen's Kappa value utilizes a columns selection method coupled with a Logistic Regression model. This approach proved effective in automatically identifying key columns during training.

We have used Logistic Regression for taking the most important columns for making accurate predictions. Before applying this model, we made the process more efficient. First, we excluded columns with low variance, making sure only dynamic columns were included. We also dealt with the problem of highly correlated columns by using a Linear Correlation calculator and a Correlation Filter. This ensured that the columns we have selected were diverse and worked well together.

Participated Columns

As the data does not contain many columns that directly represent consumer behaviour, many other columns that might represent consumer behaviour indirectly along with some columns that represent horse, to check every possible column that might influence the prediction, here are the columns participated in the selection:

- Columns might represent horse (object) behaviour:
    - Number of premiums per object
    - Race
    - Horse age (Ålder_häst)

- Columns might represent consumer behavior:
    - Salesperson: differ between customers that choose web or a certain salesperson
    - Payment term: customers, who choose to pay monthly, are different from those who choose annual or seasonally payment
    - Payment method: customers, who choose Autogiro, are different from those who choose Faktura

- Deductible Vet care: differ between customers that 25% or 50%
- Number of objects: total number of objects that a customer has
- Number of premium: total number of premiums that a customer has
- RegistrationPeriod: represent for how long the customer has been having insurance in Dustan measured in months
- RegistrationPeriodYears: represent for how long the customer has been having insurance in Dustan measured in years
- County
- Age of the client
- Gender of the client
- Number of claims: number of claims per costumer

- Columns might represent costumer behaviour were added in later iteration:
  - AvClaimsPerPersonPerYear (Average claims per person per year)
  - AvClaimsperPersonperObject (Average claims per person per object)
  - AvClaimsPerPersonPerObjectPerMonth (Average claims per person per object per month)
  - AvClaimsPerPersonPerObjectPerYear (Average claims per person per object per year)

The best result was achieved by combining RegistrationPeriod and AvClaimsperPersonperObject which is the dominated influencer with about 99%, other columns were excluded, see figure 5 below that shows the Cohen's kappa factor for excluded data were used for prediction without the dominating column. It is very clear that in best combination Cohen's kappa factor did not reach more than 0.08.

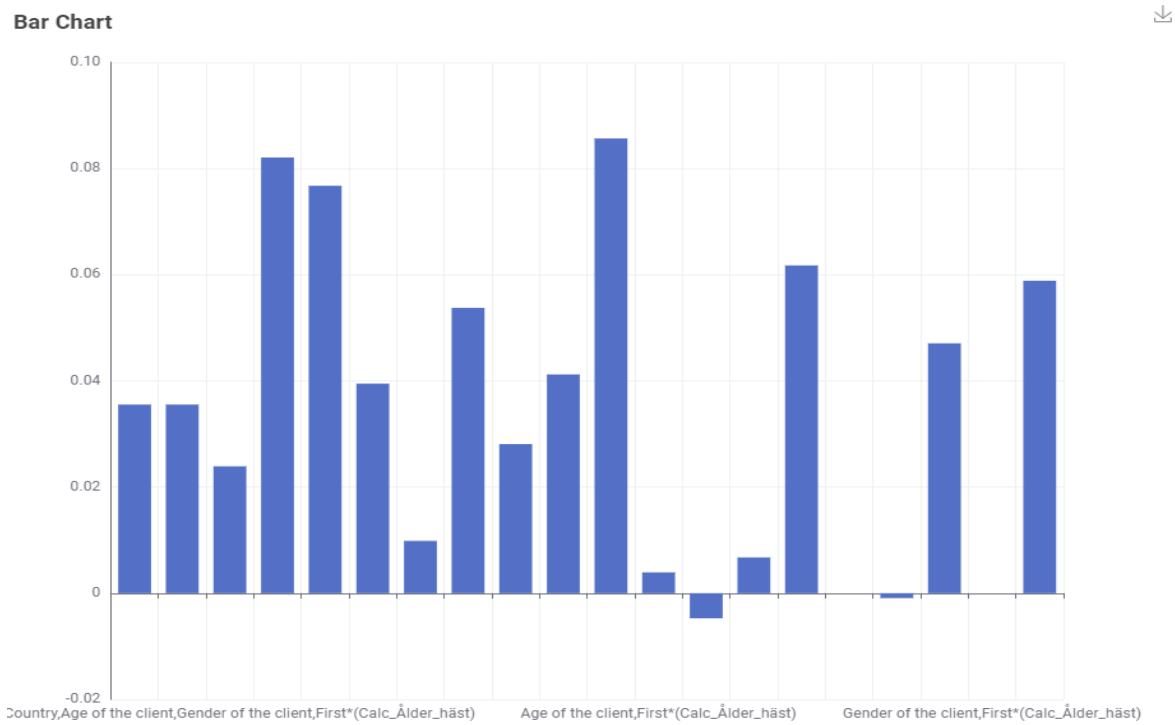

Figure [5] shows Cohen's kappa factor of combination of the rest of columns.

To ensure that the result the whole column selection process was repeated using Random Forest model instead of logistic regression achieving the same result.

Preparing the model

When the participating columns were cleaned and pre-processed, it was time to build the model. To solve the problem, a classification model was chosen to be built. The desired result that was expected from building a classification model was not only to classify if a horse(object) has a claim or not but also to calculate the probability for a horse getting at least a claim. The calculated probability can be the base to build a pricing model, based on customer behaviour in the future.

Technique

To prepare the prediction model we had use different classification models available in KNIME. The type of models is listed below:

- Decision Tree
- Random Forest
- Logistic Regression
- Support Vector Machine (SVM)
- K Nearest Neighbour
- Gradient Boosted Tree
- XGBoosted Tree
- Naïve Bayes

Testing criteria

The models were assessed using three criteria:

- Prediction accuracy: It is the percentage that defines how well the model have predicted correctly.
- Cohen's Kappa: It is a measure which ranges from -1 to 1. If the value is 1, it tells that the value is perfect agreement.
- Area under the curve: Its value lies between 0 and 1. 1 denotes that the model is able to classify into two different classes, in our case the customers who raised claim and who did not.

Training input and parameter settings

These are the dataset inputs used for training the models:

• Number of claims: number of claims per customer
• RegistrationPeriod
• HadClaim: represent if a horse had a claim or not

To get the best model we had also tried changing the parameter in each of the models and the major change we noticed is as below:

Increasing the Min Number Record Per Node from 2 to 6, increased the AUC very much.

Model assessment

Comparing all models' performance shows that the Logistic Regression classification model performed the best in terms of the Accuracy and Cohen's Kappa, see figure 6 below. However, SVM model performance is too close to Logistic Regression performance in terms of the Accuracy and Cohen's Kappa it even performs better in terms of area under ROC curve, see figure 7 below.

Row ID	D Overall Accuracy	D Overall Error	D Cohen's kappa
XGBoost Tree	0.9	0.1	0.579
K Nearest Neighbor	0.886	0.114	0.523
Random Forest	0.899	0.101	0.584
Decision Tree	0.894	0.106	0.562
Logistic Regression	0.909	0.091	0.607
SVM	0.906	0.094	0.603
Naive Bayes	0.905	0.095	0.598
Gradient Boosted	0.894	0.106	0.556

Figure [6] shows the overall accuracy and Cohen's Kappa of all models.

Row ID	D Area Under Curve
Logistic Regression	0.943
SVM	0.946

Figure [7] shows the area under the curve of all SVM and Logistic regression models.

Evaluation

Evaluation of Model′s Performance

In order to assess the effectiveness of the developed predictive model in predicting claim events and its alignment with the business objectives of the insurance company, various performance metrics were utilized. The model achieved an accuracy of 90.9%, indicating that it correctly predicted 3799 of the claim events in the dataset and 382 have been incorrectly predicted. However, accuracy alone might not provide a complete picture, especially if the dataset is not balanced.

Relating to the company's objectives, while the model shows moderate accuracy, these shows for enhancement to better align with the goal of reducing claims costs and improving profitability. Further analysis reveals potential difference in the model's predictions and actual outcomes, says the need for refinement before deployment. Recommendations for model improvements are crucial to ensure better alignment with the company's objectives and enhance its predictive capabilities for cost-saving initiatives.

However, further analysis and potential adjustments might be necessary to refine the model's predictive capabilities for more accurate and reliable predictions aligned with the insurance company's goals.

Assessment of Model′s Performance

Considering the business objective of reducing claims costs and improving profitability, the model's predictive power needs improvement, particularly in reducing incorrect results. Additionally, the insights derived from this model can aid in targeted training and education programs to prevent claims in specific areas identified by the model.

Conclusion and Recommendation

Even our model can predict claims with high accuracy which can be the base for building efficient pricing model considering the customer behavior, the model doesn’t take any consideration for the time of having claim. These issues might not satisfy the company needs has the company wants to know the probability of having a claim in a certain period which in most cases a year. Building the model based on a premium having a claim instead of a horse having a claim leads to more reliable and used for results. Building such a model was too difficult due to the lack of time and different type of movement (type of policy). To solve this problem a model required to building for each type of policy, it consumes a lot of time but the result might satisfy the business goals. Our recommendation for the company is to include some columns in their dataset represent the customer behavior efficiently e.g. year of experience that a customer has taking care of horses, education level of the customer and training level of customer.

References

Author unknown (n.d). What is the CRISP-DM methodology? Smart Vision Europe. Obtained: 2023- 12-10. https://www.sv-europe.com/crisp-dm-methodology/#one

Appendix

Table 3

Table 3 is showing the Premium data in Swedish and English.

Nr	Swedish	English
1	Utpris	Price
2	Förfallen (utpris)	Written Premium
3	Intjänat (utpris)	Earned Premium
4	Produkt	Product
5	Moment	Type of Policy - Insurance element
6	Anslutningsdatum	Date of accession
7	Förmedlare	Salesperson
8	Försäkringsnummer	Policy Number
9	Hästens namn (värde)	Name of the Horse
10	Period start	Policy Inception date
11	Period slut	Policy Expiry Date
12	Betalningstermin	Payment term
13	Förmedlare inmatad	Salesperson entered
14	Betalsätt	Payment method
15	Produkt ID	Product ID
16	Ras	Race
17	Anslutsdatum månad	Month of accession
18	Inmatad datum	Date of entry
19	Referens	Reference
20	Annullerad	Date of termination
21	Försäkringsår	UW Year (Underwriting Year)
22	Län	County
23	Försäkrad ålder	Age of the client
24	Försäkrad kön	Gender of the client
25	Försäkrad kundnr	Customer number

26	Självrisk veterinärvård	Deductible Vet care
27	Född	Year of birth (Horse)

Table 4

Table 4 is showing the Claims data in Swedish and English.

Nr	Swedish	English
1	Skadenummer	Company ID insured
2	Skadedatum	Date of Loss
3	Registrerad	Date of Notification
4	Status	Status
5	Typ	Type
6	Kund	Name of Insured (customer)
7	Diagnoskod	Type of Claim
8	Aktuell reserv	Indemnity Outstanding
9	Försäkring Självrisk..	Deductible Vet care
10	Utbetalningar	Indemnity Paid
11	Försäkringsbelopp	Sum insured
12	Total reserv	Total Incurred
13	Avslutad	Closed
14	Försäkring period start	Policy Inception Date
15	Försäkring period slut	Policy Expiry Date
16	Beslut	Decision
17	Försäkringsnr	Policy number
18	Moment	Type of Policy – insurance element
19	Försäkrinv Vet vår Värde	Deductible Vet care (value)
20	Försäkringsår	Underwriting Year
21	Avser typ	Regarding Type
22	Kund kundnr	Customer number